CSI BISHOP SOLOMON DORAISAWMY COLLEGE OF ARTS AND SCIENCE

Department of Computer Science

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Project Documentation

Complete Technical Report

Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking

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Platform	ServiceNow

Abstract

This project demonstrates the automation of employee data import and relationship mapping within ServiceNow using Import Sets, Transform Maps, and Dot Walking. The solution allows organizations to import bulk employee records from external CSV/Excel files and automatically surface related information (Department, Email, Manager) on Incident forms without manual lookup or custom scripting. The project was implemented on a ServiceNow Personal Developer Instance (PDI) as part of the Naan Mudhalvan initiative.

1. Introduction

1.1 Background

ServiceNow is a leading cloud-based ITSM platform used by organizations worldwide to manage IT services and workflows. One of its most powerful capabilities is the Import Set framework, which enables administrators to automate data migration from external systems into ServiceNow tables.

In most organizations, new employee data is maintained in HR systems or spreadsheets. Manually entering this data into ServiceNow is error-prone and time-consuming. Additionally, service desk agents handling incidents need to quickly identify an employee's department and manager — information not natively displayed on the Incident form without configuration.

1.2 Project Objectives

- Automate the import of employee records from a CSV/Excel file into ServiceNow using Import Sets.
- Configure a Transform Map to correctly map staging table fields to the sys_user (User) table.
- Implement Dot Walking on the Incident form to automatically display employee relationship data.
- Achieve all objectives using native ServiceNow tools without custom scripting.

1.3 Scope

- Employee data import via file-based Import Sets.
- Transform Map creation with Auto Match and Mapping Assist.
- Dot Walking configuration on the Incident form.
- End-to-end testing of the complete workflow.

2. Technologies Used

Technology	Purpose	Details
ServiceNow PDI	Primary development environment	Vancouver / Latest
Import Sets	Bulk data import from external files	Native ServiceNow feature
Transform Maps	Field mapping from staging to target table	Native ServiceNow feature
Dot Walking	Auto-populate related fields on forms	Native ServiceNow feature
Google Sheets	Employee data preparation	Free web-based tool
Microsoft Excel	Alternative data preparation & export	CSV/XLSX format

3. System Architecture

3.1 Tier 1 — Data Source Layer

Employee data is maintained in Google Sheets with columns: name, email id, phone number, Department, manager. The data is exported as an Excel (.xlsx) file and used as the import source.

3.2 Tier 2 — Import & Transform Layer

ServiceNow's Import Sets module is used to load the Excel file into a staging table, map fields to the sys_user table using a Transform Map, and execute the transformation to insert records.

3.3 Tier 3 — Presentation Layer

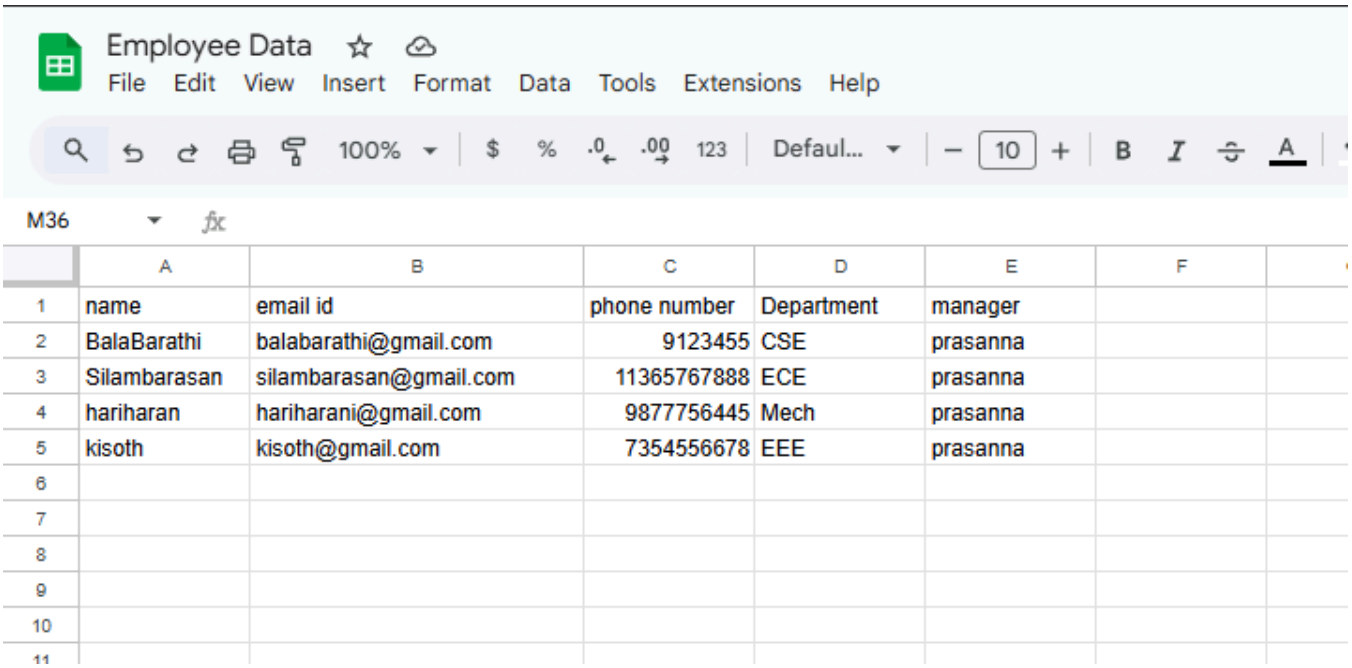
Dot Walking traverses reference fields on the Incident form to automatically display the assigned user's Department, Email, and Manager without any custom code.

4. Implementation

4.1 Milestone 1: Data Import

Step 1 — Data Preparation

A Google Sheet was created with 4 employee records: BalaBarathi (CSE), silambarasan (ECE), hariharan (Mech), kisoth (EEE) — all with manager: prasanna. Exported as emp.xlsx.

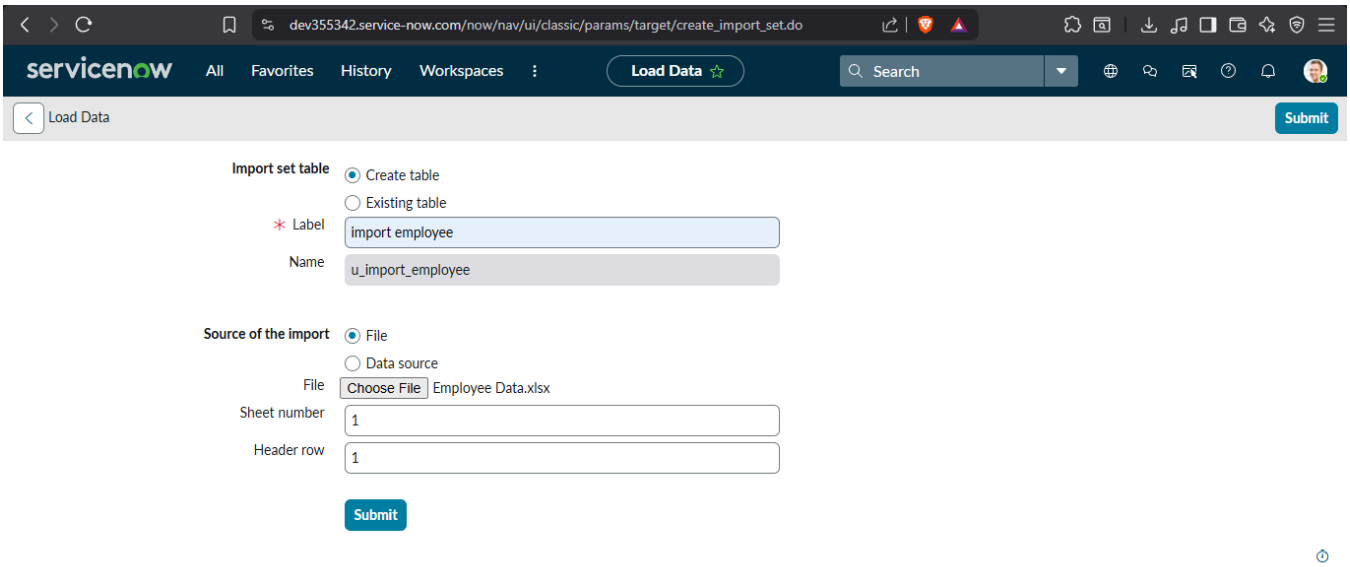


	A	B	C	D	E	F
1	name	email id	phone number	Department	manager	
2	BalaBarathi	balabarathi@gmail.com	9123455	CSE	prasanna	
3	Silambarasan	silambarasan@gmail.com	11365767888	ECE	prasanna	
4	hariharan	hariharani@gmail.com	9877756445	Mech	prasanna	
5	kisoth	kisoth@gmail.com	7354556678	EEE	prasanna	
6						
7						
8						
9						
10						

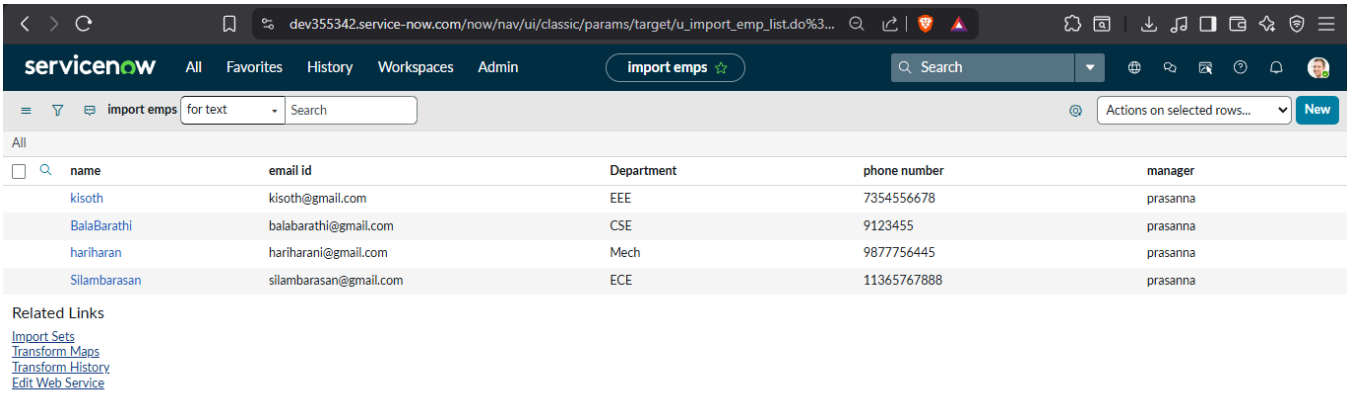
[Screenshot: Google Sheet with employee records]

Step 2 — Load Data

The file was loaded via All > System Import Sets > Load Data. A staging table was created and all 4 records were imported successfully.



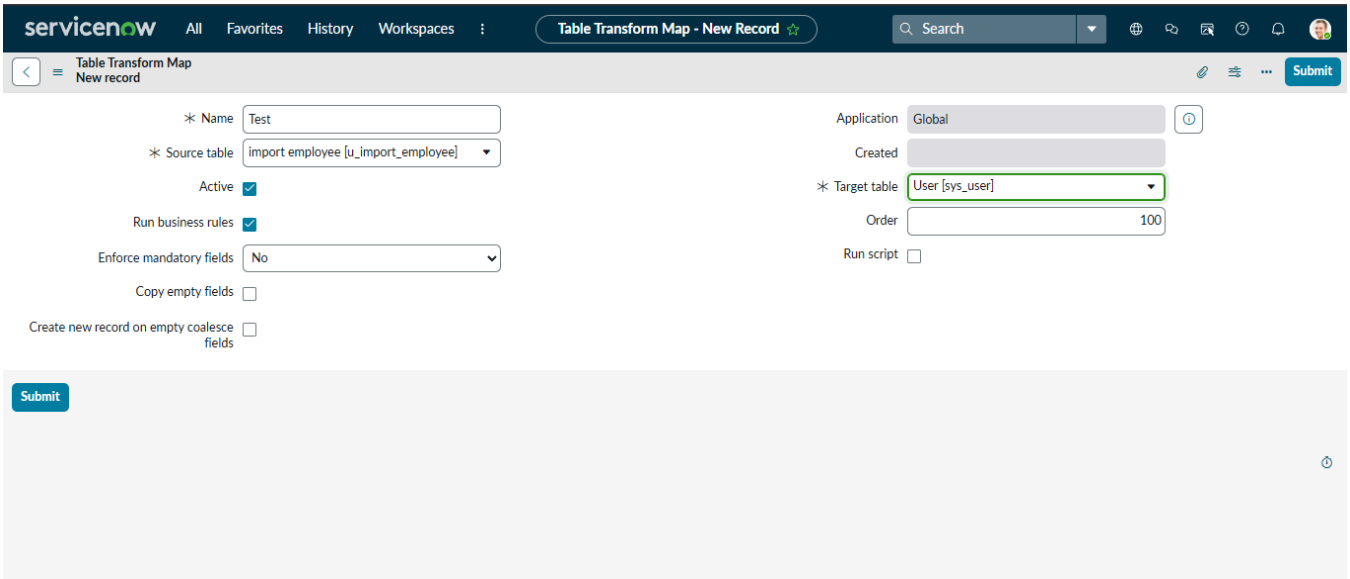
[Screenshot: ServiceNow Load Data form]



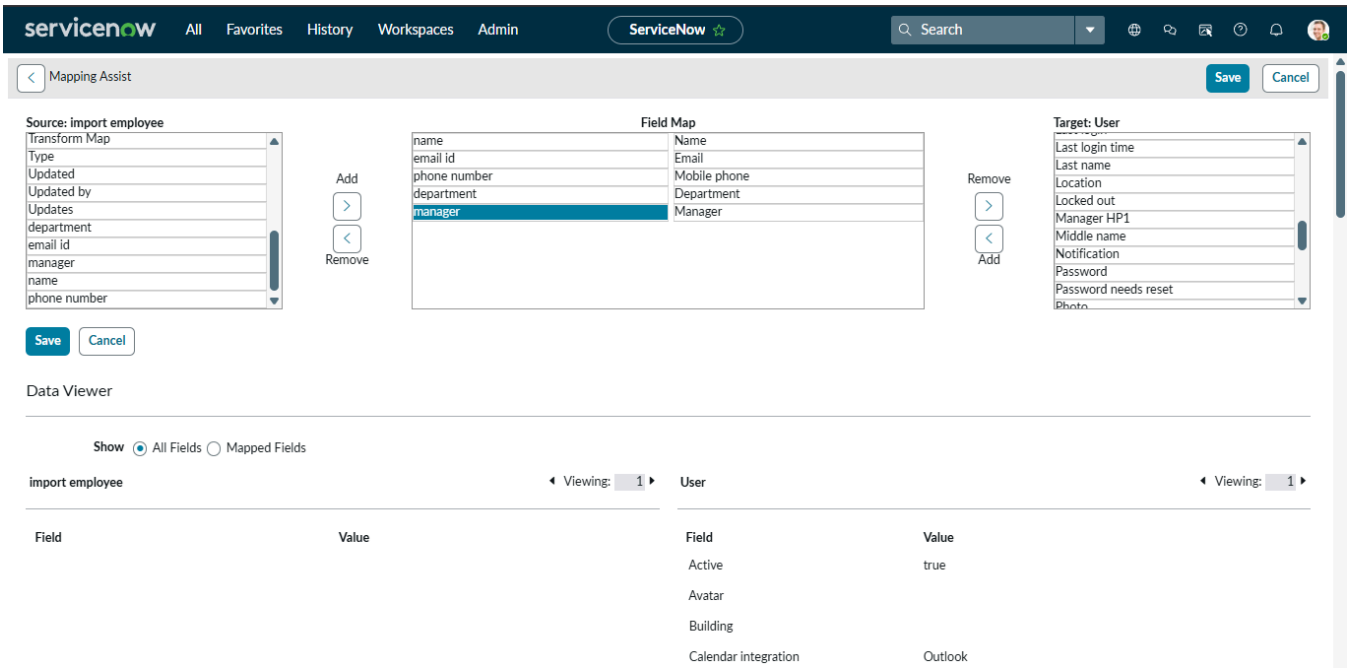
[Screenshot: Staging table with imported records]

Step 3 — Transform Map Creation

A Transform Map named 'Test' was created with source: staging table, target: User [sys_user]. Auto Map Matching Fields mapped: name, email id, phone number, Department. Manager was mapped manually using Mapping Assist.



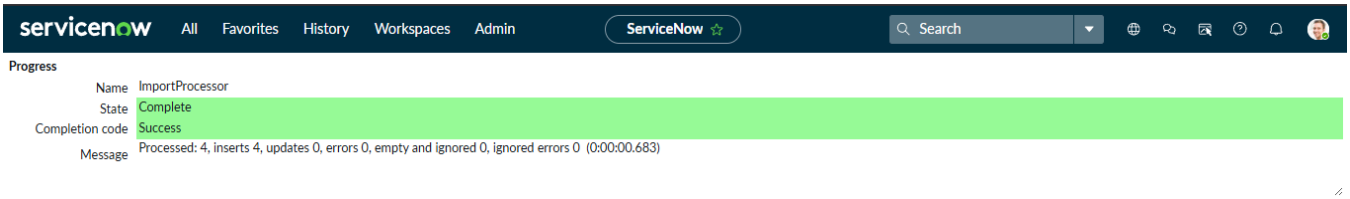
[Screenshot: Transform Map configuration form]



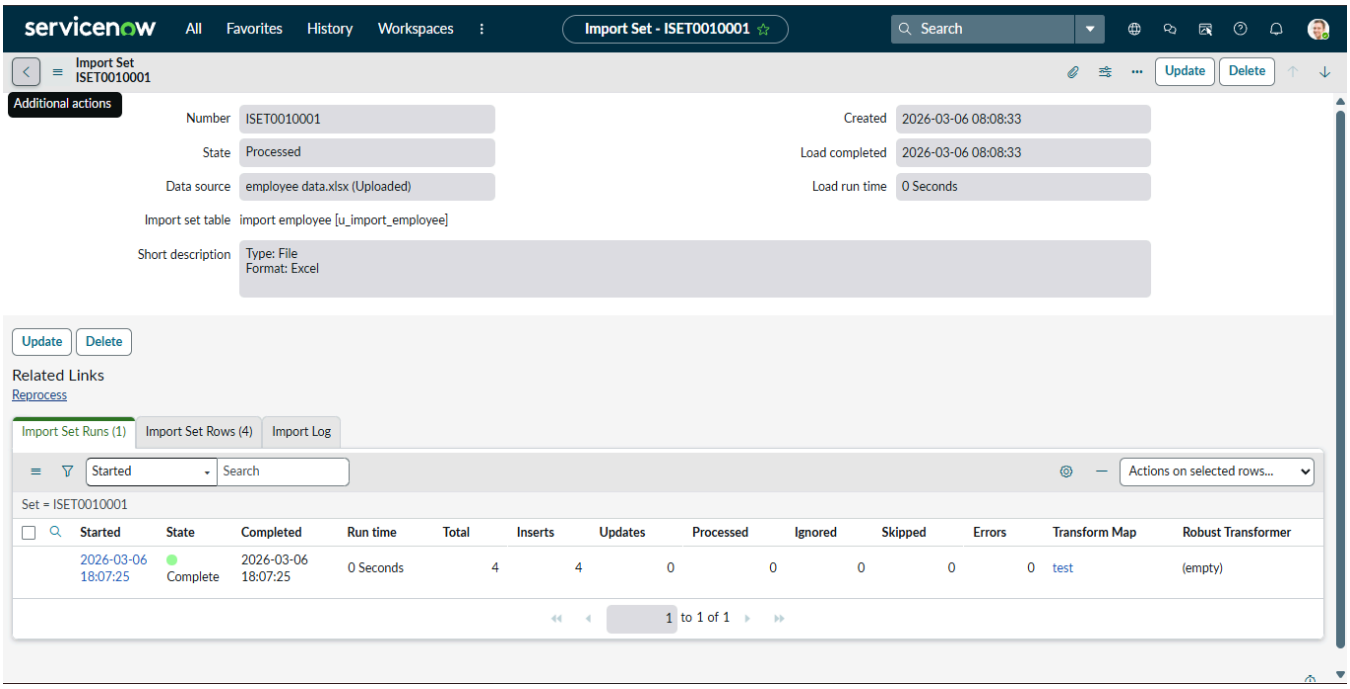
[Screenshot: Mapping Assist field mapping view]

Step 4 — Transform Execution

The Transform was executed from the Import Set record. Result: State: Complete, Completion Code: Success. Total: 4 records | Inserted: 4 | Errors: 0.



[Screenshot: Transform completion screen (State: Complete)]



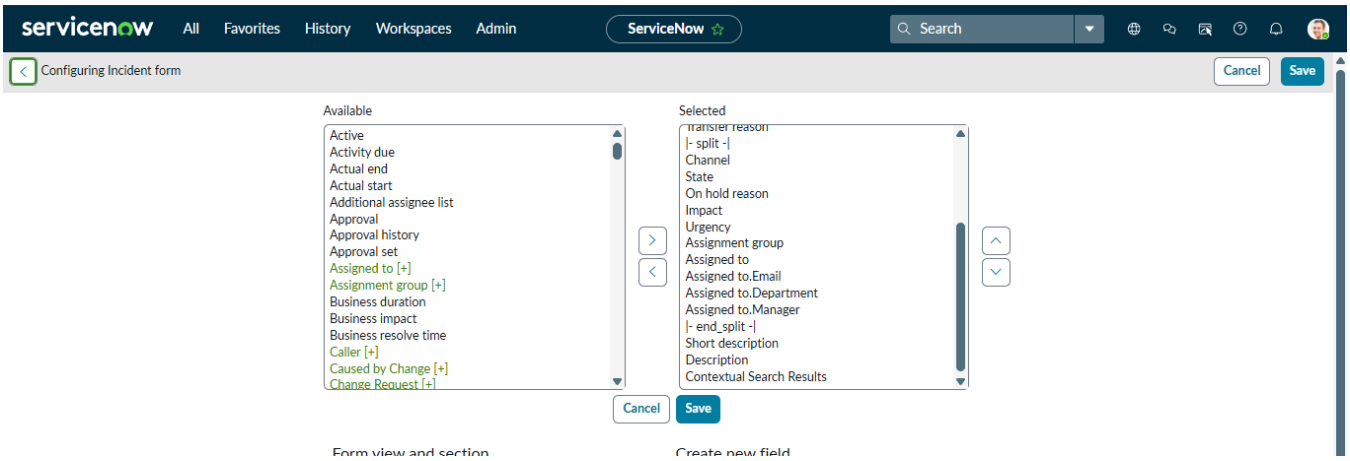
[Screenshot: Import Set Rows showing 4 Inserted records]

4.2 Milestone 2: Dot Walking

Step 5 — Form Layout Configuration

The Incident form layout was modified via Context Menu > Configure > Form Layout. Three dot-walked fields were added:

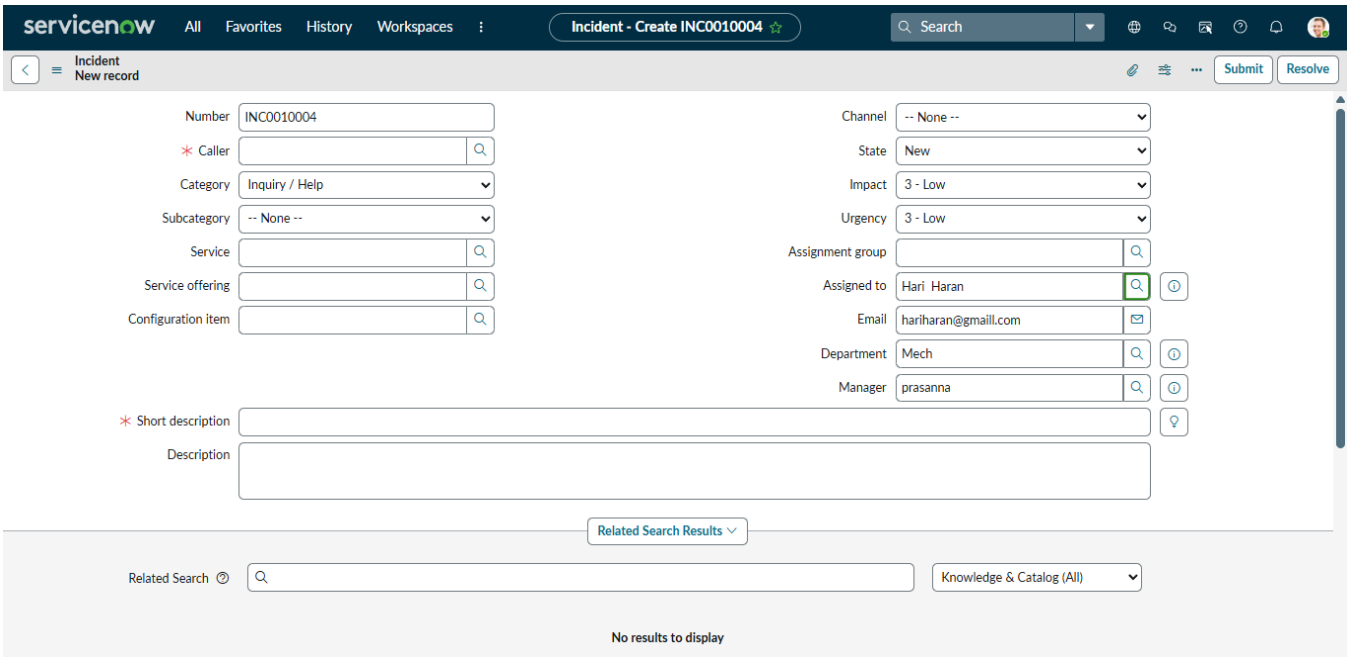
- Assigned to Department (assigned_to.department)
- Assigned to Email (assigned_to.email)
- Assigned to Manager (assigned_to.manager)



[Screenshot: Form Layout showing dot-walked fields in Selected]

Step 6 — Testing

A new Incident was created. Assigned To was set to 'mouni'. The following fields populated automatically: Email: mouni@gmail.com, Department: Mech, Manager: Shiva.



[Screenshot: Incident form with auto-populated dot-walked fields]

5. Results

#	Objective	Status	Result
1	Employee data imported from CSV/Excel file	Completed	**Pass**
2	Transform Map correctly mapped all 5 fields	Completed	**Pass**
3	4 employee records inserted into sys_user	Completed	**Pass**
4	Dot walking fields visible on Incident form	Completed	**Pass**
5	Auto-population of Dept, Email, Manager on Incident	Completed	**Pass**

6. Conclusion

This project successfully demonstrated how ServiceNow's native capabilities — Import Sets, Transform Maps, and Dot Walking — can be combined to create a fully automated employee data import and relationship mapping solution.

Import Sets provided a flexible, reusable mechanism for loading external data into ServiceNow without any custom code. Transform Maps ensured data was correctly validated and mapped to the target table. Dot Walking enabled the Incident form to automatically surface related employee information, significantly improving the experience for service desk agents.

The solution is production-ready and can be extended to support scheduled imports, additional data sources, and more complex relationship mappings.

6.1 Key Learnings

- Understanding the Import Sets framework: staging tables, transform maps, and coalesce fields.
- Using Mapping Assist for visual field mapping between source and target tables.
- Applying Dot Walking to traverse reference fields and surface related data.
- Configuring ServiceNow forms using the no-code Form Layout designer.
- Testing end-to-end data flows in ServiceNow PDI.

6.2 Future Enhancements

- Implement scheduled imports using a Data Source to automatically sync employee data daily.
- Add a coalesce field on the email column to prevent duplicate records on re-import.
- Extend dot walking to additional fields such as assigned user's location and phone number.
- Integrate with HRMS via REST API for real-time employee data synchronization.

References

- ServiceNow Documentation: Import Sets — docs.servicenow.com
- ServiceNow Documentation: Transform Maps — docs.servicenow.com
- ServiceNow Documentation: Dot Walking — docs.servicenow.com
- Naan Mudhalvan Program — naanmudhalvan.tn.gov.in
- SMARTBRIDGE Learning — smartbridge.in